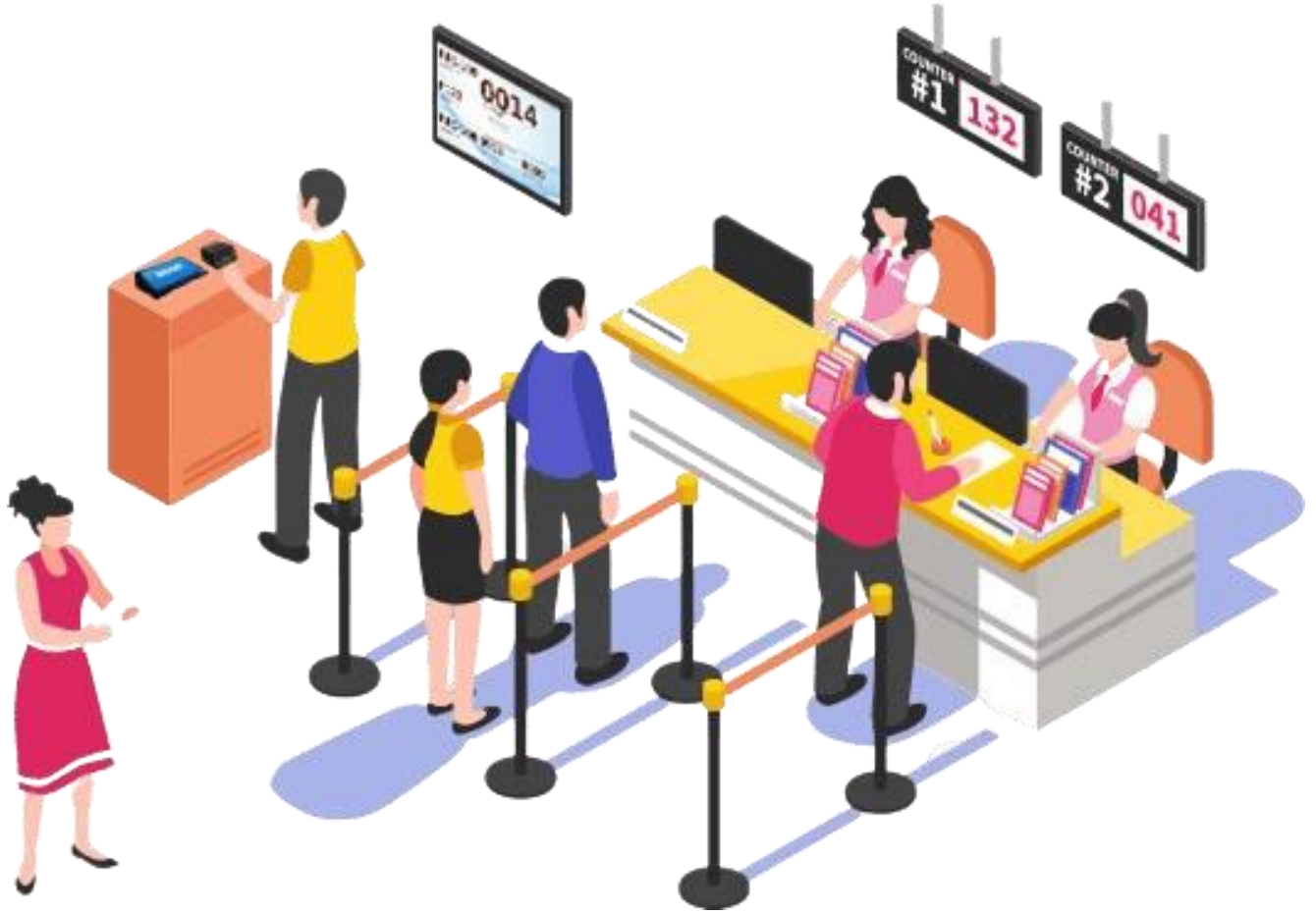
Scam lens

Queue Management System

SUB : FOAI

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ABSTRACT

QueueLess is a web-based queue management system developed to simplify queues in offices, laboratories and service centers. Users can view available services, obtain queue tokens and check their token status. The system displays waiting tokens and the people currently being served, helping users understand the queue without having to wait blindly.

View queue details

Show token details and whether a token is waiting or being served.

View services

Display available services and the number of people waiting.

Staff controls

Allow staff to call the next token and pause service when needed.



Software and Hardware Requirements

Software

- Visual Studio Code
- Node.js and npm
- Google Chrome

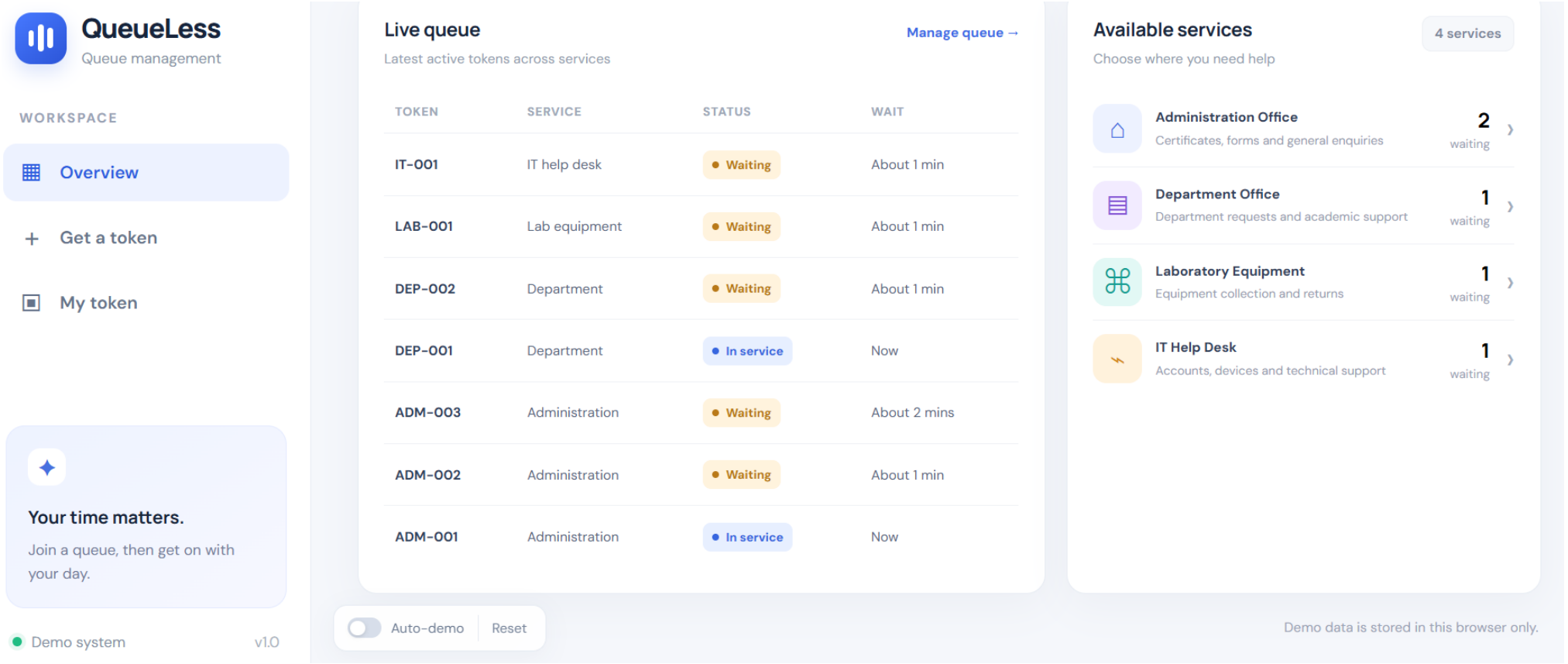
Hardware

- Intel Core i3 or equivalent
- Minimum 4 GB RAM
- At least 1 GB free storage



FULL VIEW OF THE PROJECT

The overview page shows the live queue and the services currently available to users.



SAMPLE CODING

Example logic for selecting the next waiting token.

```
function callNextToken(queue) {
  const nextIndex = queue.findIndex(
    token => token.status === "Waiting"
  );

  if (nextIndex === -1) return queue;

  return queue.map((token, index) =>
    index === nextIndex
      ? { ...token, status: "In service" }
      : token
  );
}
```

Find next token

Finds the first token with Waiting status.

Update status

Changes the selected token to In service.

ARCHITECTURE DIAGRAM

The main parts of the current browser-based prototype are shown below.

User interface

Overview, Get a token, My token and Staff console.

Queue logic

Creates and updates tokens and tracks queue status.

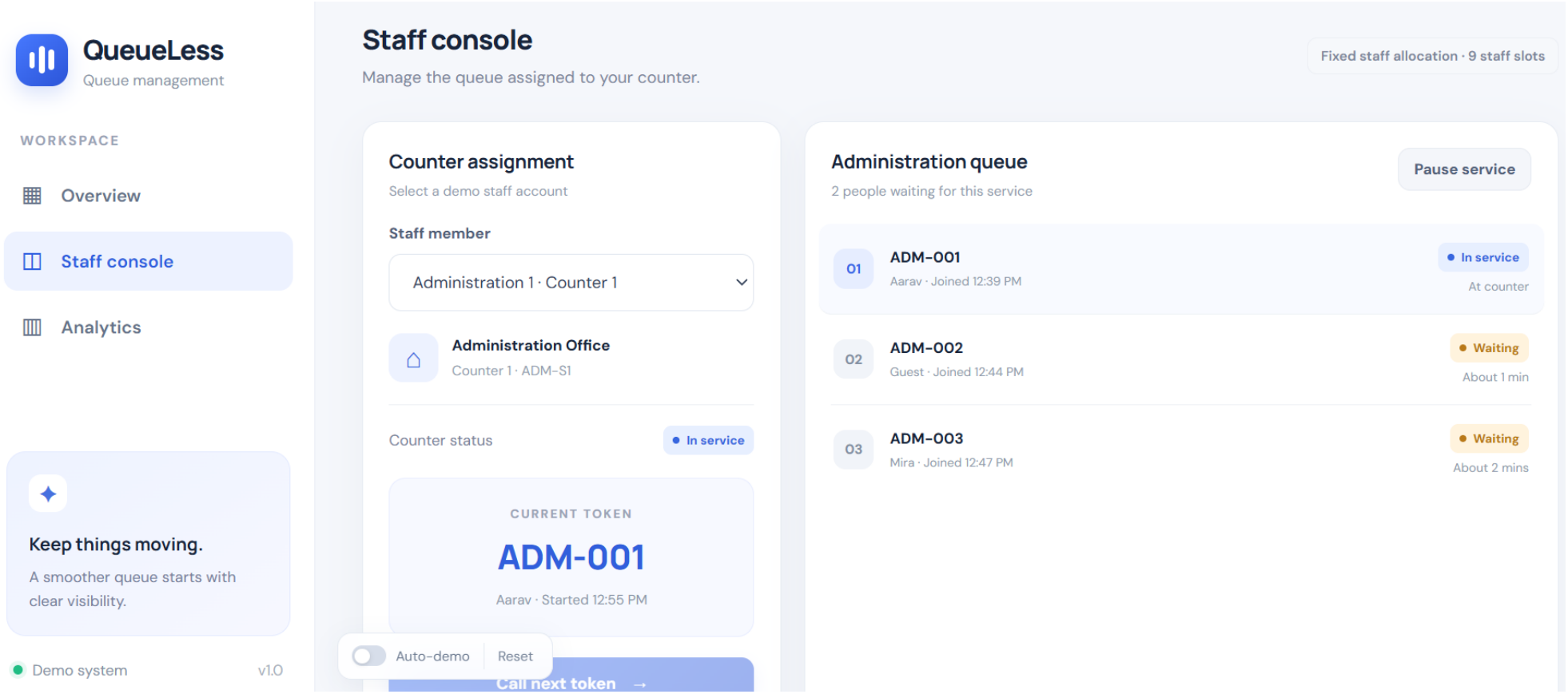
Browser storage

Stores demo data in the current browser.

The interface responds to user actions, updates queue information and keeps demo data in the browser. This version does not s how a separate server or database.

PROJECT DEMO: STAFF CONSOLE

The staff console shows the selected counter, current token and people waiting.



DEMO OF THE PROJECT

The project can be demonstrated in the following order:

- Open Overview to show the live queue and available services.
- Select Get a token and choose the required service.
- Open My token to show the token number and status.
- Go to Staff console and select a staff member or counter.
- Click Call next token and show the queue status update.

CONCLUSION

QueueLess provides a simple way to view service queues and manage tokens through a web interface.

- Users can view services and check token status.
- Staff can monitor waiting tokens and call the next person.
- Future improvements may include login, a database and live updates.

